Jimin Park

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RESEARCH INTEREST

World Model | Foundation Model | Agentic AI | Representation Learning | Neural Operator

EDUCATION

M.S. Battery Conflation Engineering | Yonsei University — Sep. 2025 – Aug. 2027

B.S. Mathematics & B.S. in Computer Science and Engineering | Dongguk University — Mar. 2017 – Aug. 2024

International Secondary Education | Shanghai Singapore International School (SSIS) — Aug. 2008 – Aug. 2014

RESEARCH EXPERIENCE

M.S. Student | Multi-Physics Modeling & Computation Lab (Prof. Jungil Choi) — Sep. 2025 – Present

Dept. of Battery Conflation Engineering, Yonsei University

Research Intern | Multi-Physics Modeling & Computation Lab (Prof. Jungil Choi) — Jul. 2025 – Aug. 2025

School of Mathematics & Computing (CSE), Yonsei University

Research Intern | Computer Vision Lab (Prof. Sangpil Kim) — Jul. 2024 – Oct. 2024

Dept. of Artificial Intelligence, Korea University

Research Intern | Computer Science & Distributed Computing Lab (Prof. Junho Jeong) — Mar. 2023 – Jun. 2024

Dept. of Computer Science and Engineering, Dongguk University

PUBLICATIONS

International Journal

1. **Park, J.M.**, Kim, K.H., and Choi, J.I., "WorldBatt: learning a world model of the lithium-ion cell for unified battery diagnostics and prognostics," manuscript in preparation for submission to *Nature Computational Science*.
2. **Park, J.M.**, and Choi, J.I., "Generative supervision for full-range state-of-health estimation from truncated aging experiments," manuscript in preparation for submission to *Applied Energy*.

International Conference

1. **Park, J.M.**, and Choi, J.I., "BMNO: A Battery Multiscale Neural Operator for Degradation Trajectory Estimation and Forecasting" *in Proc. AI for Science Conference (AI4Sci Korea),* Seoul, Korea, Sep.–Oct. 2026. ***[Oral Presentation]***
2. **Park, J.M.**, Kim, B.J., and Jeong, J.H., "An Analysis on Synthetic Data for Improving Performance of Skeleton-Based Fall Down Detection Models," *in Proc. IEEE 5th International Conference on Big Data Analytics and Practices (IBDAP)*, Bangkok, Thailand, Aug. 2024. ***[Oral Presentation]***

Domestic Conference

1. **Park, J.M.**, and Choi, J.I., "Learning Latent Dynamics as Self-supervised Pretext Tasks: Universal Battery Foundation Model for Cross-Chemistry Diagnosis and Prognosis," *in Proc. Spring Conference of the Korean Society of Automotive Engineers (KSAE),* Jeju, Korea, May 2026. ***[Oral Presentation]***
2. **Park, J.M.**, and Choi, J.I., A Universal Battery Foundation Model via Self-Supervised Learning of Latent Dynamics for Cross-Chemistry Diagnosis and Prognosis," *Spring Conference of the Korean Society for Industrial and Applied Mathematics (KSIAM*), Ulsan, Korea, May. 2026.
3. **Park, J.M.**, Kim, K.H., and Choi, J.I., "UniBatt: Universal Foundation Framework for Battery State Estimation," *in Proc. Fall Meeting of the Korean Electrochemical Society (KECS)*, Busan, Korea, Nov. 2025. ***[Oral Presentation]***
4. **Park, J.M.**, Kim, D.H., Yang, S.Y., and Seok, M.G., "Thermal Threshold-Based Pose Estimation for Smoking Behavior Detection," *in Proc. Korean Computer Congress (KCC), KIISE*, Jeju, Korea, Jun. 2024. ***[3rd Place Award]***

5. Xu, Y., Kim, B.J., **Park, J.M.**, and Jeong, J.H., "Improving Top-Down Pose Estimation-Based Fall Detection Using Segmentation," *in Proc. Spring Conference of the Korean Association of SMART Policing (KASP)*, Jeju, Korea, Jun. 2024. ***[Oral Presentation] [Best Paper Award]***
6. Kim, D.H., **Park, J.M.**, Yang, S.Y., and Seok, M.G., "Accelerating Motion Deblurring of Color Images with Grayscale Conversion and Approximate Color Restoration," *in Proc. Spring Conference of the Korea Multimedia Society (KMMS)*, Seoul, Korea, May 2024.
7. Yang, S.Y., **Park, J.M.**, Kim, D.H., and Seok, M.G., "CycleGAN-Based Image Synthesis for Improving Traffic Light Recognition in Rainy Conditions," *in Proc. Spring Conference of the Korea Multimedia Society (KMMS)*, Seoul, Korea, May 2024.
8. **Park, J.M.**, Kim, B.J., Xu, Y., and Jeong, J.H., "A Research on Synthetic Data for Improving Performance of Skeleton-Based Fall Down Detection Models," *in Proc. Spring Conference of the Korean Institute of Intelligent Systems (KIIS)*, Seoul, Korea, Apr. 2024. ***[Oral Presentation] [Best Paper Award]***

PROJECTS

Proposal Co-Author & Lead Researcher | "Development of an AI-Based Integrated Design and Safety Prediction Platform for Next-Generation Battery Materials-Electrodes-Cells", AI+S&T Innovation Technology Development (R&D) Program, National Research Foundation of Korea (NRF), Ministry of Science and ICT — April. 2026 – Dec. 2029

Proposal Author & Lead Researcher | "Universal Foundation Model for Battery Diagnostics," Advanced GPU Computing Support Program, National IT Industry Promotion Agency (NIPA), Ministry of Science and ICT — Mar. 2026 – Dec. 2026

Proposal Author & Lead Researcher | "Physics Foundation Model for Battery Full-Lifecycle Prediction and Diagnostics," Institute for AI and Social Innovation — Jan. 2026 – Dec. 2026

Proposal Author & Lead Researcher | "Development of a Foundation Model for Predicting Battery Safety and Lifetime," High-Performance Computing Support Program, National IT Industry Promotion Agency (NIPA), Ministry of Science and ICT — Sep. 2025 – Feb. 2026

Researcher | "Development of Software Technology for Predicting and Early Diagnosing for EV Battery Thermal Runaway," Ministry of Trade, Industry and Energy (MOTIE) — Jul. 2025 – Oct. 2027

Researcher | "Copyright Protection of AI Model & Digital Content from Generative AI," Collaboration with Google DeepMind — Jul. 2024 – Oct. 2024

Researcher | "Management for Police Field Customized Research and Development," Ministry of Science and Technology & Korean National Police Agency — Mar. 2024 – Jun. 2024

Researcher | "Development of SMART Community Policing System," Ministry of Science and Technology & National Research Foundation of Korea (NRF) — Mar. 2023 – Dec. 2023

HONORS & AWARDS

Selected Project | AI+S&T Innovation Technology Development (R&D) Program — 2026

Spotlight Session Award (Outstanding Poster) | "UniBatt," LG Tech Conference, LG Corporation — 2026

Selected Project | NIPA Advanced GPU Computing Support Program — 2026

Selected Project | Institute for AI and Social Innovation Support Program — 2026

Selected Project | NIPA High-Performance Computing Support Program — 2025

Best Paper Award | Korean Association of SMART Policing (KASP) Spring Conference — 2024

Best Presentation Award | Korean Association of SMART Policing (KASP) Spring Conference — 2024

The Grand Prize (President's Award) | ICIP & Capstone Design Contest, Dongguk University, *Team Leader* — 2024

Best Paper Award | Korean Institute of Intelligent Systems (KIIS) Spring Conference — 2024

3rd Place Award | Korean Computer Congress (KCC), KIISE — 2024

The Grand Prize (Dean's Award) | Adventure Design Contest, Dongguk University, *Team Leader* — 2022

The Grand Prize (President's Award) | Writing Technical Report and Presentations Contest, Dongguk University, *Team Leader* — 2022

ACTIVITIES

Researcher | "LG Tech Conference (TC)," LG Corporation, LG Science Park (Magok)	May. 2026
Attendee | "INTERBATTERY 2026," COEX, Seoul, Korea	Mar. 2026
Trainee | "KISTI AI-Powered Physics Bootcamp", NVIDIA, OpenACC organization, KISTI, Hewlett Packard Enterprise (HPE)	Feb. 2026
Team Leader | "AI-Driven Autonomous Patrol Robot Framework for Real-Time Smoking Behavior Detection in Diverse Environments", ICIP & Capstone Design, Software Education Institute at Dongguk University	Sep. 2023 – Jun. 2024
Researcher | "An Advanced Study on Data Generation for Deep Learning Analysis Using the Game Engine", Independent Capstone Design, Dongguk University	Sep. 2023 – Dec. 2024
Researcher | "A Study on Data Generation for Deep Learning Analysis Using the Game Engine", Independent Capstone Design, Dongguk University	Mar. 2023 – Jun. 2023
Developer | "Crawling and NLP-Powered Dongguk University Notification Alert System", Open Software Project, Software Education Institute, Dongguk University	Mar. 2023 – Jun. 2023
Team Leader | "Artificial Intelligence Window Air Purifier", Leaders in Industry-University Cooperation 3.0, Adventure Design, the Ministry of Education and National Research Foundation of Korea	Jun. 2022 – Dec. 2022
Team Leader | "AI Coder" Technical Report Writing and Presentation, Innovation Center for Engineering Education & College of Engineering, Dongguk University	Jun. 2022 – Dec. 2022

PATENTS

1. Choi, J.I., Kim, K.H., **Park, J.M**., and Park, S.H., "Universal Battery Foundation Model Apparatus and Method for Battery Diagnosis," Korean Patent Application No. 10-2026-0071495, filed Apr. 21, 2026. (Applicant: Yonsei University Industry-Academic Cooperation Foundation)

SKILLS

Programming: Linux, Python, C++ | **Frameworks:** PyTorch, Scikit-learn | **Tools:** Git, Docker

Languages: English (Fluent), Japanese (Intermediate), Chinese (Intermediate)